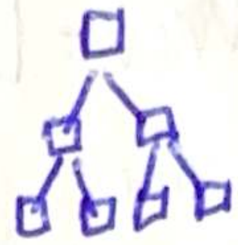
Merkle Tree



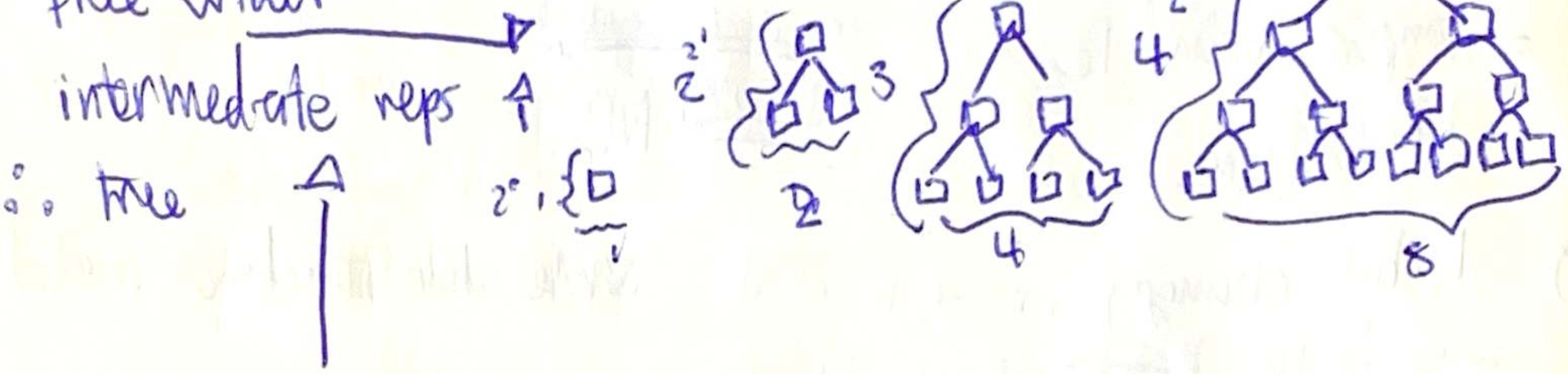
- Hash ptrs applied to a binary tree
- Blocks are the leaf nodes
- All nodes between leaf & root are intermediate representations.

∴ as data blocks ↑ ... leaf nodes ↑

tree wider

intermediate reps ↑

∴ tree



⊛ only need remember the root hash (consensus) to ensure integrity of merkle tree of transactions.

[2.3] The Advantage of MerkleTree Form: Concise Membership Proof & Non-Membership Proof.

Membership proof:

Provide: 1) data block target

2) sequence of hashes from data block to root node

... no need to store/traverse full tree

For a tree of n nodes ... a user only needs to show $\sim \log(n)$ items.

∴ $O(\log(n))$ to prove member

Non-membership proof.

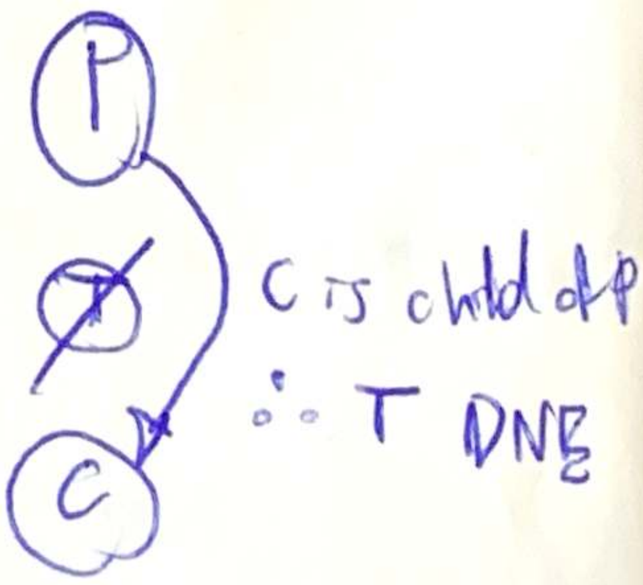
→ requires sorted merkle tree

→ show parent of target node P

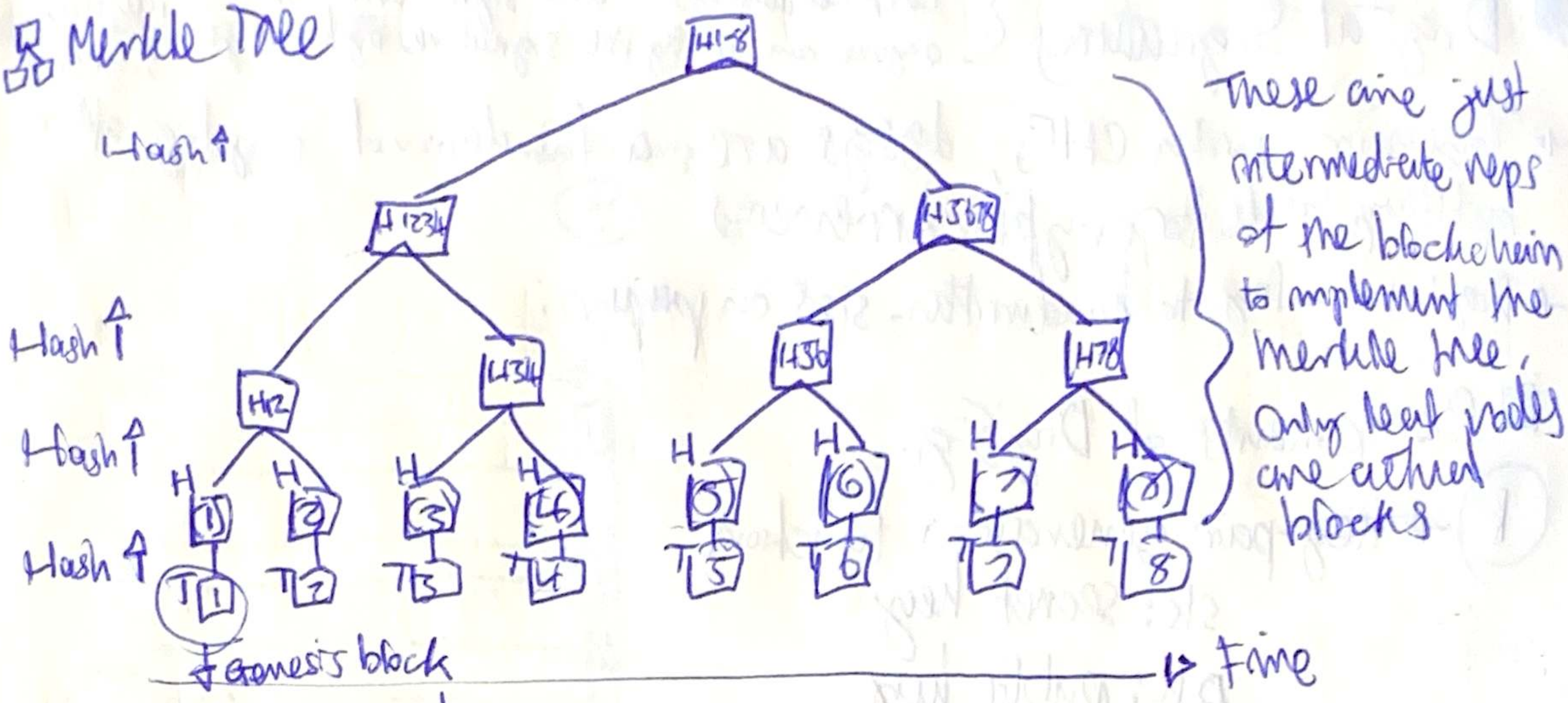
→ show child of target node C

→ Prove C is child of P

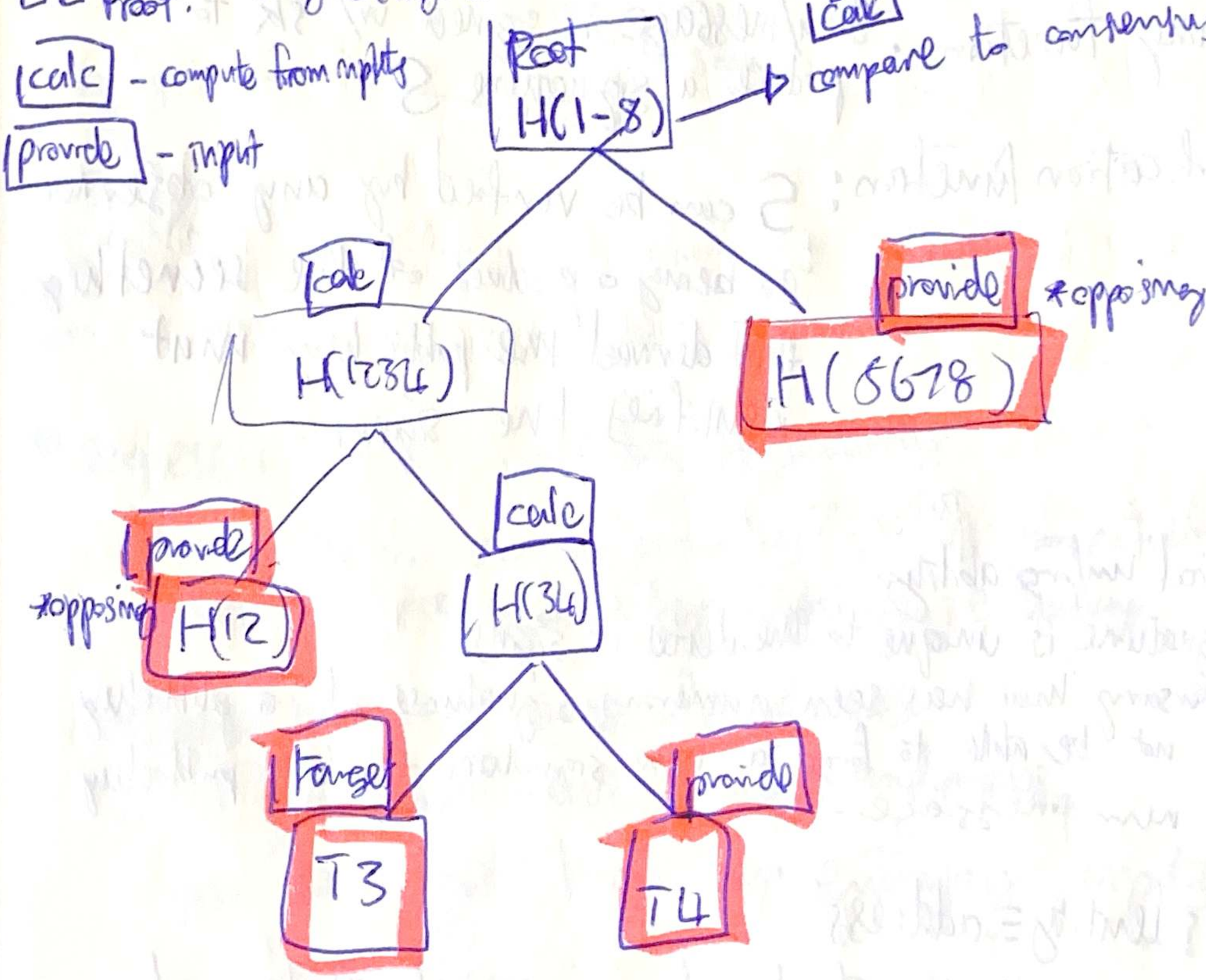
∴ no target between



Merkle Tree



Membership You don't need the tree!
Proof. only O(log(N)) of tree!



Merkle trees must be DAGs: directed acyclic graphs.
→ if not, hash computation is recursively infinite

